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A novel MILP model for the operation of the poultry supply chain

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Abstract: This work introduces a new MILP mathematical model for the poultry supply chain, which simultaneously addresses two main tasks: (1) the assignment and scheduling of the vehicles devoted to transport poultry from breeding farms to the slaughterhouse, and (2) the scheduling of the operations at the slaughterhouse. A set of reduced size instances is solved with CPLEX to illustrate the results obtained, although the model complexity prevents from obtaining feasible solutions in some cases.

Keywords: Poultry supply chain, resource assignment and scheduling, Mixed-Integer Linear Programming.

1. INTRODUCTION

In modern supply chains, integrating manufacturing systems with transportation networks plays a crucial role in synchronizing raw material and services handling, production processes, and product distribution. Semi-finished goods are transported between facilities for further processing (Lee and Chen, 2001), while finished products are delivered to customers, warehouses or distribution centers by cargo vehicles (do C. Martins et al., 2021). There is extensive literature that studies the distribution of products from production plants to other facilities, such as plants, clients, warehouses, or distribution centers.

In addition, the delivery of raw materials or services to production plants involves coordinating sourcing, procurement, and transportation to ensure timely and accurate deliveries. This process involves managing inventory levels and scheduling deliveries to optimize timing and quantities, ensuring that materials are delivered on time and in the correct sequence to prevent delays, and assigning resources by allocating machines, labor, and vehicles to tasks to enhance performances and minimize idle times (Suh et al., 1997; Koulouris and Kotelida, 2011; Jachimczyk and Myhan, 2022).

The above-mentioned tasks of assignment and scheduling are generally carried out through the formulation of decision-making problems, not only in industrial applications, but also in the framework of agricultural holdings. For instance, for crop production, machinery is assigned to tasks like land preparation; in poultry harvesting, vehicles are dispatched to farms to collect poultry and transport them to slaughterhouses and meat processing plants; in forestry, transportation resources are used to deliver timber to sawmills. In all these cases, resources (particularly vehicles) are to be continuously reassigned based on task completion: after completing a task and returning to their origin point, resources become available for reassignment to new tasks as needed. This approach aims to ensure the timely fulfillment of customer orders, reduce lead times, and optimize resource utilization (Li et al., 2023). Note that, in some cases, these applications are somehow related to the classical Vehicle Routing Problem (VRP) (Lin et al., 2023; Hashemi-Amiri et al., 2023), although this is not always possible due to the specific features of farming operation.

The present work addresses the case of the operation of poultry production supply chain. This problem falls within the scope of assignment and scheduling problems, as it is required to process a specific number of poultry daily to meet the demands of multiple customers. Accordingly, the vehicles, initially available at the processing plant (slaughterhouse), are to be assigned to breeding farms, where available poultry has to be harvested and transported back to the slaughterhouse. This process involves determining which vehicle is assigned to each breeding farm, as well as establishing the timing and sequence of departures from the processing plant to the breeding farms. Several key factors must be taken into account during this process, including the care of the animals, resource availability and compliance with regulatory requirements concerning fasting periods and animal welfare. After harvesting, the

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poultry are placed in crates, specially designed cages that ensure safe and efficient transportation. It is worth mentioning that the problem tackled here does not match the conditions of the VRP, since vehicles only harvest one farm at a time, and go back to the slaughterhouse before heading to another farm. In addition, the operation of harvesting vehicles should be programmed in relation with the slaughterhouse schedule, so that both sub-problems are intricately connected and cannot be treated separately. Therefore, an ad-hoc solution strategy accounting for both aspects of the whole problem must be designed.

Related works include different aspects of the poultry supply chain. For instance, E. Brevik (2020) proposed a mathematical model that focuses on the planning of the different steps from eggs selection from breeders, hatchery period during which eggs are incubated, breeding in broiler farms and finally, slaughter dates. The objective consists in minimizing total costs, including egg discarding costs, unhatched egg costs and boiler shortage at farms; as well as penalty costs for not fulfilling broiler client demands and the target weight at slaughterhouses. Unlike the work presented here, the catching team schedules to harvest broiler and transport them to the slaughterhouse is only marginally tackled. Due to complexity of the resulting MILP model, three rolling time-horizon heuristics are developed to provide high quality solutions for a Norwegian broiler production company.

Similarly, N. Praseeratasang (2019) introduced a nonlinear mathematical model that formulates the problem as a multi-period assignment problem: at each period, broilers from different breeding farms are to be assigned to production plants to fulfill their demand, while vehicles have to be assigned to each harvest performed at breeding farms. No operation scheduling is carried out in this case. The model is solved with LINGO and an Adaptive Large Neighborhood Search technique, for the case study of farm-plant-vehicle assignments in the Buriram Province in Thailand.

Also, A.L. Solano-Blanco (2023) proposed an integrated production planning for the broiler chicken supply chain, with growth uncertainty. First, a two-stage stochastic model that, under a set of growth scenarios, determines a minimum cost plan of orders and sizes, as well as order modifications for each scenario to meet a specified target weight at slaughterhouses. Then, once this order planning is obtained, a MILP model is formulated for the cheapest allocation of chicken lots to breeding farms, considering their capacity and biosecurity conditions. This two-steps methodology is applied to a case study in Santa Marta, Colombia.

N. Mostaghim (2024) adopted a broader point of view on the design and planning of the poultry meat supply chain, considering not only the breeding farms and slaughterhouses echelons, but also packing and processing facilities as well as distribution centers and retailers. No operation concepts are tackled, but rather decisions regarding facility locations, and flows among the different actors of the supply chain. In addition, several aspects are also included: perishability of the chicken meat products is accounted for, with its associated costs; resilience to uncertain events such as unexpected disruptions of the production line, addressed through a two-stage stochastic programming approach; two objectives are optimized simultaneously through goal programming, namely minimization of the CO_2 emissions from transportation processes and maximization of the overall profit. The resulting computational tool is applied to several provinces in Iran, obtaining significant benefits in terms of robustness and reliability of the supply chain.

Despite the limited related works, to the best of our knowledge, no study in the existing literature considers the continuous reassignment of resources, particularly vehicles, under specific conditions such as poultry regulations on fasting periods and animal welfare, which are treated as hard constraints that must be respected. This paper contributes by addressing this gap, proposing a mathematical model that incorporates all particular assumptions of the problem into the assignment and scheduling of resources to minimize the makespan – the total time required to complete all tasks – while ensuring compliance with regulatory requirements.

The structure of the paper is as follows. Section 2 provides a formal and detailed description of the problem under study. The mixed integer mathematical model of the problem is presented in Section 3. In Section 4, we present the results of numerical experiments to evaluate the performance of the proposed model and discuss the findings. Finally, Section 5 concludes the paper and outlines potential directions for future research.

2. PROBLEM DESCRIPTION

This study addresses the scheduling and logistics of a poultry processing system involving a central slaughterhouse (denoted as PB), a set I of breeding farms and a set K of homogeneous vehicles. Each breeding farm $i \in I$ has predefined demand that specifies the required number of trips by vehicles for the collection of poultry. These demand values guide the logistics and scheduling process, dictating how many trips are needed to fulfill the total requirement from each breeding farm i. It is assumed that the predefined demands from each breeding farm are satisfied by utilizing the maximum capacity of each vehicle $k \in K$. Each vehicle k departs from the processing plant (slaughterhouse), PB, to collect for the collection of poultry from a designated breeding farm i. The travel to breeding farm i requires TR_i minutes, while the harvesting process at each breeding farm i takes a fixed duration of $TCOSE$ minutes. After loading the crates, vehicle k returns to PB in TR_i minutes, where crates are unloaded (requiring TD minutes), followed by a slaughtering process lasting TS minutes. To ensure animal welfare and processing efficiency, poultry must not exceed a maximum waiting time TMH, referred to as the maximum allowed hangar time, in the holding crates before being sacrificed. Each vehicle k then undergoes a TL-minute sanitation cycle, preparing it for the next tour. It is worth clarifying that resources are limited during the unloading and sanitation cycle phases of the vehicles. While carrying out these tasks, the resources cannot be assigned to any other activities. Each vehicle k is permitted a maximum set S of collection tours, where each tour $s \in S$ limits the vehicle to a single breeding farm visit. It is important to note that each vehicle, on each tour, visits only one farm due to a health constraint. Therefore, the poultry harvested at a farm must be transported to the slaughtering plant for processing before beginning any other harvesting. In other words, a vehicle can transport poultry from only one farm per trip.

The objective is to coordinate the vehicle assignments in poultry harvesting, dispatching, and scheduling tasks, including the scheduling of breeding farm visits and the slaughter sequence, in a manner that minimizes the makespan. The makespan, in this context, refers to the total time needed to complete all tasks, from the first vehicle's departure to the final processing at PB. The goal is to ensure that all vehicles perform the assigned tasks as efficiently as possible while adhering to operational constraints (e.g.,maximum allowable hanger time), thereby reducing delays and ensuring optimal throughput.

3. MIXED INTEGER LINEAR PROGRAMMING MODEL

In this section, we propose a MILP model for the problem. Firstly, the notation (sets and parameters) are defined, thence, the decision variables, objective functions and constraints sets.

Sets:

- I: breeding farms $\{1, ..., |I|\}$
- K: vehicles $\{1, ..., |K|\}$
- S: tours of vehicles $\{1, ..., |S|\}$
- B: triplet of possible combinations of breeding farms, vehicles, and tours of vehicles $\{(i, k, s)|i \in I, k \in K, s \in S\}$

Parameters:

- $STPB$: starting time of labor in PB
- HT: harvesting time of poultry at breeding farm
- TD: discharging time of poultry in PB
- TR_i: traveling time from breeding farm $i \in I$ to PB.
- TS: time for sacrifice of poultry [min]
- TV_i: traveling time from PB to breeding farm $i \in I$
- TMH: maximum time of waiting of poultry in the PB before sacrifice
- TL: washing time of a vehicle after delivering poultry to PB
- M: very large positive number

Decision variables:

- $Z_{i,k,s}$: binary variable that takes the value of 1 if vehicle $k \in K$ pick up the harvested poultry of breeding farm $i \in I$ in its tour number $s \in S$
- $SEQ_{i,k,s,j,u,l}$: binary variable, equal to 1 if i-th farm's poultry, harvested by vehicle k in its tour s, is slaughtered immediately before j-th farm's poultry, harvested by vehicle u in its tour l $\forall (i, k, s), (j, u, l) \in B \cup \{(0, 0, 0)\}$, 0 otherwise. $(0, 0, 0)$ is a dummy triplet for tour's starting and ending.
- $TI_{i,k,s}$: departure time of vehicle $k \in K$ in its tour number $s \in S$ to the breeding farm $i \in I$
- $TSV_{i,k,s}$: departure time of vehicle $k \in K$ in its tour number $s \in S$ from the breeding farm $i \in I$
- $TIS_{i,k,s}$: starting time of sacrifice of poultry of breeding farm $i \in I$ delivered by vehicle $k \in K$ in its tour number $s \in S$
- $Cmax$: Makespan

Objective function:

$$Min \qquad Z = Cmax \tag{1}$$

s.t.:

$$\sum_{k \in K, s \in S} Z_{i,k,s} = 1 \qquad \forall i \in I \tag{2}$$

$$\sum_{i \in I, i \neq j} Z_{i,k,s-1} \geq Z_{j,k,s} \qquad \forall j \in I, \forall k \in K, \forall s \in S, s > 1 \tag{3}$$

$$\sum_{i \in I} Z_{i,k,s} \leq 1 \qquad \forall k \in K, \forall s \in S \tag{4}$$

$$TI_{i,k,s} \leq M \cdot Z_{i,k,s} \qquad \forall i \in I, \forall k \in K, \forall s \in S \tag{5}$$

$$TSV_{i,k,s-1} + TD + TR_i + 10 \leq TI_{j,k,s} + M \cdot (2 - Z_{i,k,s-1} + Z_{j,k,s}) \qquad \forall i \in I, \forall j \in I, \forall k \in K, \forall s \in S, s > 1 \tag{6}$$

$$TSV_{i,k,s} \geq HT - M \cdot (1 - Z_{i,k,s}) \qquad \forall i \in I, \forall k \in K, \forall s \in S \tag{7}$$

$$TIS_{i,k,s} \geq TSV_{i,k,s} + (TD + TR_i) \cdot Z_{i,k,s} \qquad \forall i \in I, \forall k \in K, \forall s \in S \tag{8}$$

$$TIS_{i,k,s} \leq M \cdot Z_{i,k,s} \qquad \forall i \in I, \forall k \in K, \forall s \in S \tag{9}$$

$$TIS_{i,k,s} \geq (STPB + TD) \cdot Z_{i,k,s} \qquad \forall i \in I, \forall k \in K, \forall s \in S \tag{10}$$

$$TIS_{i,k,s} - (TSV_{i,k,s} + (TD + TR_i) \cdot Z_{i,k,s}) \leq TMH \cdot Z_{i,k,s} \qquad \forall i \in I, \forall k \in K, \forall s \in S \tag{11}$$

$$TIS_{j,u,l} \geq TIS_{i,k,s} + TS \cdot Z_{i,k,s} - M \cdot (1 - SEQ_{i,k,s,j,u,l}) \qquad \forall (i, k, s) \in B, \forall (j, u, l) \in B \tag{12}$$

$$TIS_{j,u,l} >= STPB - M \cdot (1 - SEQ_{0,0,0,j,u,l}) \qquad \forall (j, u, l) \in B \tag{13}$$

$$\sum_{(j,u,l) \in B} SEQ_{0,0,0,j,u,l} = 1 \tag{14}$$

$$\sum_{\substack{(i,k,s)\in B\cup\{(0,0,0)\},\\ i\neq j}} SEQ_{i,k,s,j,u,l} = Z_{j,u,l}$$

$$\forall (j,u,l) \in B \quad (15)$$

$$\sum_{\substack{(j,u,l)\in B,\\ i\neq j}} SEQ_{i,k,s,j,u,l} \leq Z_{i,k,s} \qquad \forall (i,k,s) \in B \quad (16)$$

$$SEQ_{i,k,s,i,k,s} = 0 \qquad \forall (i,k,s) \in B \quad (17)$$

$$\sum_{\substack{(i,k,s)\in B\cup\{(0,0,0)\},\\ i\neq j}} SEQ_{i,k,s,j,u,l} = \sum_{\substack{(i,k,s)\in B\cup(0,0,0),\\ i\neq j}} SEQ_{j,u,l,i,k,s} \qquad \forall (j,u,l) \in B \quad (18)$$

$$TIS_{i,k,s} + TS \cdot Z_{i,k,s} \geq TIS_{j,u,l} + TS \cdot Z_{j,u,l} - M \cdot (1 - SEQ_{i,k,s,0,0,0})$$

$$\forall (i,k,s) \in B, \forall (j,u,l) \in B, i \neq j \quad (19)$$

$$TI_{i,k,s} + (TV_i + HT) \cdot Z_{i,k,s} = TSV_{i,k,s}$$

$$\forall i \in I, \forall k \in K, \forall s \in S \quad (20)$$

$$TSV_{i,k,s} + (TD + TR_i + 10) \cdot Z_{i,k,s} - TI_{j,k,h} \leq 960 + M \cdot (2 - Z_{i,k,s} - Z_{j,k,h})$$

$$\forall i \in I, \forall k \in K, \forall s \in S, \forall h \in S, s \geq h, s \leq |S| \quad (21)$$

$$Cmax \geq TIS_{i,k,s} + TS \cdot Z_{i,k,s}$$

$$\forall i \in I, \forall k \in K, \forall s \in S \quad (22)$$

The objective function (1) minimizes the makespan, i.e., the final time of the last scheduled operation, as an indicator of the process efficiency. The constraints establish a structured process for scheduling tasks and vehicle assignments in a poultry harvesting and slaughter sequence. According to constraint set (2), each harvest task i is assigned to exactly one vehicle k in a specific tour s. To maintain order in these assignments, constraint set (3) specifies that a new tour for any vehicle k can only be scheduled after completing the previous one. Constraint set (4) further ensures that each vehicle k in a tour s is assigned to at most one task i, maintaining a one-to-one task allocation. The timing of tasks is also carefully regulated. Constraint set (5) states that if vehicle k is not assigned to a task i in tour s, no start time for that task is recorded. Constraint set (6) enforces a sequence constraint for each vehicle, ensuring that one tour ends before the next begins for the same vehicle. Constraint set (7) establishes that the departure time is the arrival time plus the harvest duration. Constraint set (8) ensures that the starting time of the slaughtering process is greater than or equal to the vehicle's arrival time at the slaughtering plant. Constraint set (9) prevents any start time for slaughter if the binary variable $Z_{i,k,s}$ is zero. Constraint set (10) ensures that if assignment variable $Z_{i,k,s}$ takes the value of 1, the slaughter begins only after the start of operations at the sacrifice plant. To avoid delays, constraint set (11) enforces a maximum wait time for poultry in the hangar, while constraint set (12) indicates that the starting time of a sacrifice is greater than or equal to the starting time of the immediately preceding sacrifice plus the sacrifice duration. The waiting time for the first vehicle arriving at the sacrifice plant is outlined in constraint set (13), which applies if the vehicle arrives after operations have commenced. To manage the slaughter sequence, constraint set (14) ensures that only one vehicle arrives first at the sacrifice plant, requiring the introduction of a fictitious task, vehicle, and tour labeled (0,0,0). Constraint set (15) mandates that all vehicles collecting chickens must arrive at the sacrifice plant, while constraint set (16) ensures that each task completed by a vehicle at the sacrifice plant has at most one unique predecessor. Furthermore, constraint set (17) guarantees that a vehicle is neither its own predecessor nor successor. Sequence and precedence constraints among tasks and vehicles in tours are enforced by constraint set (18). Constraint set (19) designates a final task whose slaughter end time must be equal to or later than all prior tasks, ensuring the orderly completion of the entire sequence.

Constraint set (20) ensures that a vehicle leave a breeding farm after its arrival to that breeding farm. Constraint set (21) enforces a maximum labor time of 960 minutes for each vehicle. Finally, constraint set (22) calculates the makespan.

4. RESULTS OF MILP MODEL

Seven instances were randomly generated based on the locations of real breeding farms, with one instance for each breeding farm quantity from 8 to 14 breeding farms. For each instance, the model was executed with configurations of 2 and 3 vehicles, resulting in a total of 14 different runs. The results of the MILP model, in terms of $C_{\max}$, execution time, and *relmipgap*, are reported in Table 1.

As observed, the complexity of the problem increases with the number of breeding farms. For instances with 13 or more breeding farms and 3 vehicles, no feasible solution was found within 27,000 seconds of execution time. Similarly, for instances with only 2 vehicles, no feasible solution was found for instances with 11 or more breeding farms. These results also indicate that using fewer vehicles makes it more challenging to satisfy the constraint of a maximum working time of 960 minutes per vehicle. Furthermore, only for three instances with 8 and 9 breeding farms was the model able to obtain the optimal solution in less than 27,000 seconds. Finally, Table 1 also presents the *relmipgap*, which measures the difference between the best integer (or feasible) solution found so far and the best possible solution, defined as the theoretical lower bound for minimization or the upper bound for maximization. The average *relmipgap* for the instances that found a solution within 7 and a half hours of execution was 10.57%. However, it can be seen that this value increases as the instance size grows, further confirming the complex nature of the problem.

Table 1. MILP results

Breeding farms	Vehicles	Makespan	Execution Time (s)	relpmipgap
8	2	1559.4	930.274	0.00
8	3	1397.4	9825.816	0.00
9	2	1678.2	12060.685	0.00
9	3	1460.6	27000	11.34
10	2	*	27000	*
10	3	1554.4	27000	16.69
11	2	*	27000	*
11	3	1701.2	27000	23.88
12	2	*	27000	*
12	3	1661.4	27000	22.06
13	2	*	27000	*
13	3	*	27000	*
14	2	*	27000	*
14	3	*	27000	*

* No feasible solution found after 27000s of execution.

5. CONCLUSION

In this work, a mathematical model is proposed for the assignment and scheduling of resources in the framework of the poultry supply chain. The mathematical model incorporates considerations of animal welfare by ensuring that the waiting time for poultry does not exceed a certain threshold, while also optimizing the scheduling of breeding farm visits and the slaughter sequence to minimize the time required for all operations. The results obtained on some randomly generated instances show that small size cases can be optimally solved. However, as soon as the instance size grows, no feasible solution can be found, therefore justifying the development of an ad-hoc heuristic solution procedure, which may constitute a perspective for future works.

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